Segment-Focused Portfolio Optimization Based on Machine Learning Predictions

Marek Šugár*

Supervisor: Samuel Rosa[1]†

Department of Applied Mathematics and Statistics, FMFI UK, Mlynská dolina, 842 48 Bratislava

Abstract: In this paper, we present a complex analysis of approaches to portfolio optimization based on predictions made by machine learning (ML) algorithms. We compare the prediction accuracy of several ML algorithms based on a unified sample of feature variables. In the paper, we focus on the segment of information technology (IT) companies listed on the NYSE and NASDAQ exchanges and use data from September 2017 to January 2026. Based on the results of our analyses, we propose a portfolio optimization strategy that combines the Markowitz mean-variance approach with ML predictions.

Keywords: quantitative finance, portfolio optimization, machine learning, stock price prediction, stock segment

1 Introduction

The development of asset prices in financial markets is a complex and non-linear time series which is affected by a large number of external factors [Yin et al., 2025]. Having the ability to approximately predict the course of asset prices in the future is of great significance to investors. Confident price forecasting can be used to devise informed investment strategies and for capital optimization.

This paper studies the possibility of devising a unified data pipeline capable of generating stock price predictions and implementing them in the optimization of capital allocation over a given historical time interval.

Among a number of studies involving the forecasting of stock prices, many tend to shift their focus to a very limited number of stocks [Huang, 2024, Yin et al., 2025] or to mainstream stock indices [Krauss et al., 2017]. There is a lack of studies conducted on coherent groups of stock assets. What is more, is the fact that these papers tend to shift focus solely on prediction results and mechanics and do not employ them in further application analyses, which are potentially of great importance for real-world investment decision-making.

The paper focuses on a specific stock market segment. Among the 500 largest companies listed on stock exchanges in the USA, by market capitalization, a total of 65 belong to the **Information Technology** (IT) sector. The main aim is to explore and evaluate the predictive power of selected machine learning (ML) frameworks in predicting next-day closing prices within the selected IT segment.

Consequently, the acquired predictions will be used as a foundation for building an investment portfolio consisting of stocks from the given segment.

In conclusion, the paper implements and analyzes the performance of a unified pipeline employing data pre-processing of a unified set of features related to the stock market, day-to-day closing price prediction, and, lastly, portfolio optimization. From a broader perspective, the paper presents a study of the predictability of an entire market segment in the US stock market.

2 Features used in predictions

The number of variables that are directly or indirectly generated through day-to-day market operations is vast, and one should consider subsetting them for ML purposes.

In this section, we aim to present and justify the process of our general feature selection, which is mostly based on *a priori* knowledge of processes within financial markets and the potential effects of these variables on stock markets.

For the sake of completeness, the data were obtained via an API directly fetching data available on Yahoo Finance and accessible through the Python abstraction provided by the `yfinance` library [Yahoo Finance, 2025, Aroussi, 2025].

*sugar11@uniba.sk

†samuel.rosa@fmph.uniba.sk

2.1 Stock market assets

As possible predictors for modeling, other market assets might be valuable. In the paper, we employed a selected number of **stock indices** – an abstract concept that accumulates information from various stocks into one unified asset. Indices are specifically designed to replicate the price development of a given set of stocks. By choosing well-designed indices that replicate a broad scope of the market, the model might acquire valuable predictor information.

In the study, the **S&P 500** and **NASDAQ Composite** (NASDAQ) indices were used as predictors. The S&P 500 reflects the performance of the 500 largest US-traded stocks in terms of total market capitalization. Unlike the S&P 500, the NASDAQ reflects the performance of technology and IT stocks traded in the USA, which is presumed to be beneficial, given the scope of the study.

Furthermore, one of the most fundamental parts of the market are **commodities**. This term primarily represents raw goods that are traded on markets in large quantities.

Their fundamental role in the market stems from the fact that most of these assets play an essential part in further processing, consumption, and industry. Matsumoto et al. [Matsumoto et al., 2023] suggest that the commodity market plays a vital role in the entire financial segment; therefore, including such assets in the feature set might be beneficial.

Among a large number of commodities, we chose to include data regarding **gold** and **crude oil**. Gold is traditionally considered a safe-haven asset and is often used as a hedge against economic uncertainty [Osita, 2025]. Crude oil plays a vital role in keeping the world in motion, and the majority of the world's activities depend on its availability. This suggests that these dependencies within the global market are projected into the price movements of the respective commodities.

Lastly, in order to reflect the international competitiveness of the US economy relative to other major economies and to provide quantitative information about relations among major economies (which are most likely important predictors for stock price movements), we decided to include data from the **foreign exchange market** (FOREX) – a fully decentralized market reflecting the relative value of a particular currency against another.

Given our focus on the US market, we chose to include currency pairs where the US dollar (USD) is the base or quote currency. In particular, the included pairs are the euro, Japanese yen, British pound sterling, Swiss franc, Australian dollar, Canadian dollar, and New Zealand dollar.

2.2 Other market indicators

In order to track the performance of the US economy more precisely, it is advantageous to include information regarding national debt as potential features. For this purpose, we included data on **US Treasury bonds**. According to Kumar et al. [Kumar et al., 2021], bonds of various maturities are valuable market predictors; consequently, we selected a wide range of bond maturities for the feature set. We included the 13-week maturity (short-term) and the 5-year Treasury yields (long-term).

Apart from that, we included the **Volatility Index** (VIX), which is a quantitative metric reflecting volatility of the whole US economy and serves as a gauge tracking the state of the economy in terms of risk.

3 Prediction methodologies

3.1 Rolling window

In order to properly process time series data into training, validation, and test sets, many approaches have been proposed [Diebold, 2006]. For the purposes of the **next-day prediction** employed in this paper, we adopted concepts of **rolling window** approach.

In our approach, we define a static time series interval that serves as a hybrid training-validation set. The interval is between T_1 and T_2 time points (with $T_2 - T_1 = M$). Within this interval reside two frames of variables.

Multivariate $\mathbf{X} \in \mathbb{R}^{M \times F}$, where F denotes the number of features used, serves as the input variable. Included features in $\mathbf{X}$ are values of selected financial variables, shifted backward up to $q = 9$ days. That means that $\mathbf{X}[t]$ (the t-th row of $\mathbf{X}$) represents data available from days $t-1, t-2, \ldots, t-9$.

Secondly, univariate $\mathbf{Y} \in \mathbb{R}^M$ accumulates values of the target variable (next-day closing price) corresponding to the respective rows of $\mathbf{X}$. That means that $Y[t]$ (the t-th value in the Y) is the closing price on day t.

Furthermore, we define parameter $1 \leq p < M$ (training window length parameter), which is the length of the training subwindow, indicating how many next-day predictions the model would be trained on (see Algorithm 1).

With respect to the adopted data preparation and the concepts of rolling window methodology, it is possible to generalize the process of train and validation sets split. Algorithm 1 shows which parts of the global $\mathbf{X}, \mathbf{Y}$ are directly available at the certain step of prediction.

Algorithm 1 Rolling-window on (T_1, T_2) interval

```
# The prediction target day is t + p + 1
# We have available data up to day t + p
Set look-back window length p
Retrieve the feature variables' values into the X
for each stock in database do
    Retrieve closing prices into the target Y
    for t = T_1 to T_2 − (p + 1) do
        X_train ← X[t : t + p]
        Y_train ← Y[t : t + p]
        X_predict ← X[t + p + 1]
        # X_predict the latest available predictor data
        Train model f̂ on (X_train, Y_train)
        Generate prediction: f̂(X_predict)
    end for
end for
```

In this paper, the time interval between September 15, 2017, and December 31, 2022 (1,333 trading days) is used as training-validation interval and time interval between January 1, 2023, and January 16, 2026 (761 trading days) is used as test interval.

3.2 ML frameworks

In this paper, various ML algorithms were employed. Taking into account the rather short forecast horizon, regularized variants of linear regression provide a suitable framework. Longer forecast periods tend to be nonlinear, and their patterns are harder to generalize. Consequently, we chose to employ the **Lasso** and **Ridge** regression models.

Both models are built on the premise of **least-squares estimation** with a penalty term, penalizing relatively large values of coefficients in regression. Let $\mathbf{X}$ be the design matrix, $\mathbf{Y}$ the vector of target variables, and $\hat{\boldsymbol{\beta}}$ the vector of estimated regression parameters. The fit of the Lasso model is attained via the optimization problem given by:

$$\arg\min_{\hat{\boldsymbol{\beta}}} (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}})^\top (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) + \lambda \|\hat{\boldsymbol{\beta}}\|_1, \quad (1)$$

where $\lambda \geq 0$ is the so-called tuning parameter and $\|\hat{\boldsymbol{\beta}}\|_1 = \sum_{i=1}^{p} |\hat{\boldsymbol{\beta}}_i|$ is the L1 norm.

Likewise, the Ridge regression is given as follows:

$$\arg\min_{\hat{\boldsymbol{\beta}}} (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}})^\top (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) + \lambda \|\hat{\boldsymbol{\beta}}\|_2^2, \quad (2)$$

where $\lambda \geq 0$ is again a tuning parameter and $\|\hat{\boldsymbol{\beta}}\|_2^2 = \sum_{i=1}^{p} \hat{\boldsymbol{\beta}}_i^2$ is the squared L2 norm.

The core property of these algorithms is their inability to be time-aware; that is, they cannot capture temporal dependencies by default. One of the most widely used frameworks capable of addressing this is the **Autoregressive Integrated Moving Average** (ARIMA) model. For a detailed overview of the theory underlying this model, see [Diebold, 2006]. For the overview, subsequent analyses and hypertuning it is necessary to be aware of three parameters p, d, q. Parameter $p \in \mathbb{N}_0$ represents the number of past time points, on which prediction is dependent. Next, parameter d denotes the order of **differencing**. Lastly, parameter q denotes the number of past errors incorporated in improving future predictions.

Lastly, we considered k-Nearest Neigbours (k-NN) and Random Forest Regression (RFA) algorithms in the study as well.

These five modeling frameworks were used as the predictive engine for all the prediction analyses in this paper.

Apart from these models we define **naïve benchmark** model. This framework is based on Fama's suggestion which presumes that stock price development corresponds to the random walk [Fama, 1970]. The benchmark predicts the next day closing price to be equal to the latest known closing price.

4 Predictive analyses

This section presents the obtained modeling results. For selected employed frameworks, an in-depth parameter tuning process and performance evaluation on both the training-validation and test sets are reported.

4.1 Evaluation metrics

Based on the methodology introduced in Algorithm 1, suppose we obtain a prediction for day $t+1$ (of a

certain stock) denoted as $\hat{Y}_{t+1}$. In order to measure the relative deviation from the actual value Y_{t+1}, the metric of **Absolute Percentage Error** (APE) is used. The particular APE is given as:

$$\mathrm{APE}(\hat{Y}_{t+1}, Y_{t+1}) = \frac{|Y_{t+1} - \hat{Y}_{t+1}|}{Y_{t+1}}.$$

Further, the average value of the APE for an individual stock over the entire considered time interval, is evaluated (further referred to as **Mean Absolute Percentage Error** – MAPE)

Together with the naïve prediction model, we quantify significance of prediction as a proportion of predictions made by the non-naïve model that are closer or equal (in terms of APE) to the actual closing price than naïve predictions. Further, this will be referred to as the **model dominance**.

4.2 Lasso regression

The Lasso regularization model is explicitly defined by two parameters: the previously introduced p (the length of the training window) and the regularization parameter λ, which controls the penalty in the optimization.

Figure 1 visualizes the relationship between MAPE and the selected values of parameters λ and p for the sample stock Enphase Energy Inc. (ticker: ENPH).

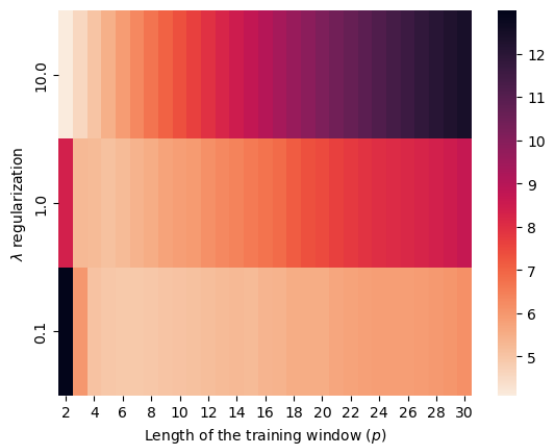

Figure 1: MAPE dependence on the rolling window training length parameter and regularization parameter (Lasso, stock ENPH).

In terms of the parameter p, it can be observed that the lowest MAPE values are reported in the interval of $p \in [4, 8]$. Based on further analyses, we chose $p = 4$ as a constant value. With this value of the parameter p, the optimal (in terms of all stocks) interval for regularization parameter was identified as $\lambda \in [4, 5]$. Figure 2 shows an overview of MAPE for the chosen ENPH stock as a function of λ.

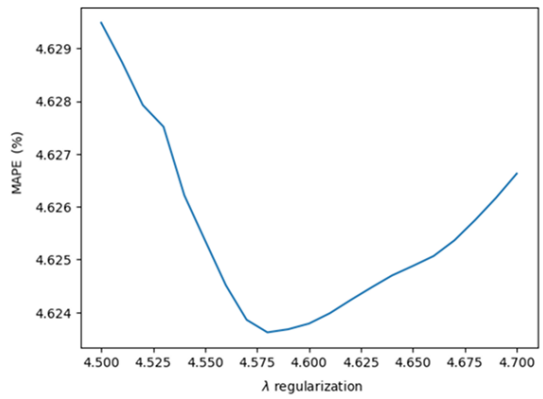

Figure 2: MAPE dependence on the rolling window training length parameter (Lasso, stock ENPH).

Such a convex pattern was observed for all analyzed stocks, indicating that the localization of the optimal value of λ was straightforward in all cases.

In the evaluation of the Lasso framework, we decided to implement two different approaches. First, we configured the Lasso model with one universal value of $\overline{\lambda}$, being the arithmetic mean of all optimized values of λ for individual stocks. The resulting value is $\overline{\lambda} = 4.15$ (referred to as the General model). Second, the model with a variable value of λ is based on identifying stock and applying its optimized value of λ in model (referred to as the Stock-wise model).

The evaluation of these models on the training-validation interval is shown in Table 1.

Model Type	MAPE (%)	Dominance (%)
General model	2.297	36.16
Stock-wise model	2.27	35.81

Table 1: MAPE and dominance values for different settings of the model (Lasso).

An interesting aspect to note is the ordering of the evaluation metrics for both models. It is possible to observe a higher value of MAPE for the General model; however, this configuration achieves comparatively better dominance. Such behaviour may appear counterintuitive, as higher proportion of predictions being closer to the actual values of the target variable might yield lower overall deviation.

4.3 Ridge regression

Since the model has the same parameters as in the case of Lasso regression, it is possible to employ the same methodology of hyperparameter tuning.

Figure 3 shows the combined dependence of MAPE on the rolling window parameter p and the regularization parameter λ (stock ENPH). Similar to the Lasso, this pattern was observed for the vast majority of the analyzed stocks.

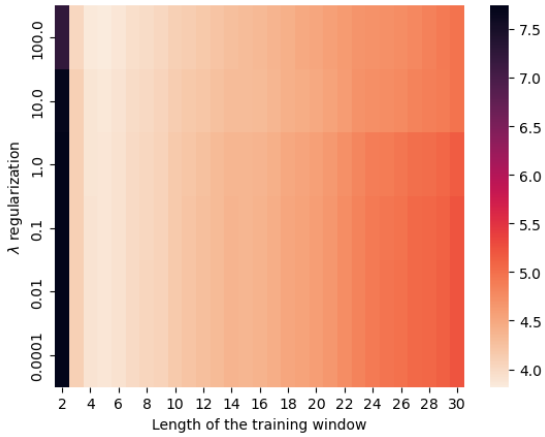

Figure 3: MAPE dependence on the rolling window training length parameter and regularization parameter (Ridge, stock ENPH).

In direct comparison with Figure 1, a clear segment of near-optimal values can likewise be observed for $p \in [4,6]$. Based on the further experiments with model fitting, we chose the value $p = 4$.

Furthermore, the dependence of MAPE solely on the λ regularization parameter can be analyzed. Figure 4 again shows a convex pattern for the ENPH stock.

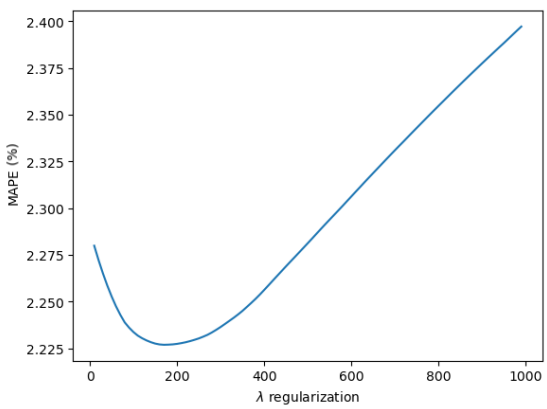

Figure 4: MAPE dependence on the rolling window training length parameter (Ridge, stock ENPH).

Even in this case, this pattern was observed for all analyzed stocks; therefore, localizing the optimal value for each stock was straightforward.

Lastly, for model evaluation, we employed the same approach as in the Lasso framework. The average value of the optimal λ across all analyzed stocks is $\overline{\lambda} = 211.85$ (referred to as the General model). The approach of using the stock-specific optimal value of λ was also preserved (referred to as the Stock-wise model). The evaluation results are presented in Table 2.

Model Type	MAPE (%)	Dominance (%)
General model	1.791	45.08
Stock-wise model	1.79	45.15

Table 2: MAPE and dominance values for different settings of the model (Ridge).

Compared to the results in Table 1, the ordering of the models is preserved – the Stock-wise model outperforms the General model in both metrics. Furthermore, the relative differences between the metric values of the models are smaller compared to those in the Lasso framework.

In direct comparison, we observe a significant improvement over the Lasso model. Although the framework is inferior to the naïve benchmark in terms of dominance, its use demonstrates measurable progress.

4.4 ARIMA model

In the case of ARIMA, we employed a similar approach to hyperparameter tuning compared to the previous two frameworks. For each rolling window, the initial state $p = 0, d = 0, q = 0$ is set. Incrementally, the algorithm explores the universe of possible settings by exploring the *neighbouring* states in the grid. All possible combinations of integer settings are fitted on the given training data, and Bayesian information criterion (BIC) is evaluated [Smith, 2026].

Subsequently, the optimal setting based on BIC is chosen. This hypertuning method is similar to the earlier hypertuning done on the regularized regression models – model is evaluated with various parameter settings on training data and the most suitable setting is chosen.

This means, there is no need to present any dependence of MAPE or dominance on the chosen model parameters. For each rolling window and for each

stock within it, the optimal settings of p, d, q are chosen based on BIC and there is no need to optimize parameters separately or visualize dependencies similar to those in previous sections.

Finally, in terms of evaluation, this set model achieved an average MAPE **1.897**% and dominance of **58.25**%. ARIMA was able to outperform all of the analyzed frameworks in terms of model dominance. On the other hand, in terms of MAPE across stocks, the Ridge model achieved better performance. Such result is particularly interesting, as a comparatively lower MAPE does not necessarily imply higher dominance.

What is more, the dominance result might arise from the fact that in approximately 42.8% of rolling windows across stocks, the p, d, q optimization terminated in a pathological case where $(p, d, q) = (0, 1, 0)$, which is equivalent to the proposed naïve benchmark method.

4.5 Other prediction frameworks

Apart from the above mentioned and analyzed frameworks, we employed other frameworks as well. Using the similar hypertuning process, k-Nearest neigbours (k-NN) and Random Forest regression (RFA) were optimized and used. Both methods are non-parametric ML methods, capable of capturing non-linear patterns, which might be the essential factor in the price prediction.

As a tuning process we employed grid-search of possible model settings and subsequent evaluation of MAPE acquired from the training-validation interval. Quite a similar phenomenon was observed in both models.

Firstly, in k-NN model, the analysis involved evaluation of models with settings given by parameter p (length of the training window) and k (number of neigbours considered). In this case, model performed the same, regardless of the value of p. The only determining factor across all stocks and all rolling window steps was the parameter k. What is more, in all stocks, across the whole training-validation interval, the optimal value was $k = 1$. On top of that, the *nearest* neigbour in all analyzed cases was the latest known value of closing price. The $k = 1$ setting actually implictly stands for the above-mentioned naïve prediction benchmark.

Moving on to the RFA framework, the number of possible settings is bigger as the number of considered parameters is larger. Hypertuning showed that the most important one is the parameter p (length of the training window), as the other parameters (e.g., depth of trees, number of trees in a forest, etc.) significantly fluctuated (with no convergence) and across all the stocks, no clear generalization was possible.

However, the optimal value of the parameter p was uniformal across all the considered stocks. The optimal value was $p = 1$. This means, that forest (and its single trees) are trained on only a single time point (the latest one). Again, the hypertuning process implicitly yields naïve model as the optimal setting.

Summing up all the considered prediction frameworks, the overview of performance on the training-validation interval is shown in Table 3.

Model Type	MAPE (%)	Dominance (%)
Naïve, k-NN, RFA	1.69	—
ARIMA	1.897	58.25
Ridge (Stock-wise model)	1.79	45.15
Ridge (General model)	1.791	45.08
Lasso (General model)	2.297	36.16
Lasso (Stock-wise model)	2.27	35.81

Table 3: Evaluation of tuned models on training-validation interval.

4.6 Test evaluation

The last step in the complex predictive analysis is evaluating performance of optimized models on previously unseen (test) data sets. Evaluation results for all the considered model variants are shown in Table 4.

Model Type	MAPE (%)	Dominance (%)
Naïve, k-NN, RFA	1.72	—
ARIMA	1.923	56.68
Ridge (Stock-wise model)	1.794	45.94
Ridge (General model)	1.798	45.88
Lasso (General model)	2.24	38.83
Lasso (Stock-wise model)	2.31	37.1

Table 4: Evaluation of tuned models on test interval.

The performance of the models tends to reach similar values as on the training-validation interval – possibly indicating no overfitting. Similar values of metrics indicate that models perform relatively stably on both (differently wide) data intervals.

5 Portfolio management

With the obtained models, which are capable of producing reliable predictions for the analyzed universe of stocks, the next step is to employ these models in portfolio optimization. In this paper, a portfolio is defined as a decision-making process yielding the allocation of capital across particular stocks in order to achieve various objectives (e.g., maximum return, minimal risk, etc.).

5.1 Theoretical foundations

Allocating capital across a set of stocks to balance the trade-off between asset volatility and portfolio return can be formulated as an optimization problem. There are several approaches to properly constructing a portfolio strategy, each requiring different types of information from the data. The foundations of the adopted frameworks and concepts in this section are defined according to Palomar [Palomar, 2025].

Taking into account the availability of next-day return predictions as a suitable baseline **Markowitz mean-variance portfolio** (MVP) was chosen. However, its elementary form tends to be impractical in practice.

Its augmented form, introducing transaction costs, was adopted. The complete definition of the used framework is as follows:

$$\max_{\mathbf{w}_t} \quad \mathbf{w}_t^\top \boldsymbol{\mu}_t - \lambda_{\mathrm{R}} \mathbf{w}_t^\top \boldsymbol{\Sigma} \mathbf{w}_t - \lambda_T \|\mathbf{w}_t - \mathbf{w}_{t-1}\|_1$$

$$\text{Subject to:} \quad \mathbf{w}_t \geq \mathbf{0}_a \text{ and } \mathbf{1}_a^\top \mathbf{w}_t = 1,$$

where the variable $\mathbf{w}_t, \mathbf{w}_{t-1} \in \mathbb{R}^a$, with a being the total number of assets, are vectors of weights of particular stock assets in the portfolio on days t and $t-1$, respectively. Further, $\boldsymbol{\Sigma} \in \mathbb{R}^{a \times a}$ is a covariance matrix of assets, λ_R is the risk-aversion hyperparameter, and λ_T is the unit transaction cost parameter. Lastly, for $\mathbf{x} \in \mathbb{R}^t$, $\|\mathbf{x}\|_1 = \sum_{i=1}^t \|\mathbf{x}_i\|$ is the L1 or Manhattan norm.

Such an extended MVP framework not only balances the return and variance of the possible portfolio, but also penalizes high volume of transactions, which in practice implies lower losses.

This allocation framework is used for daily re-allocations, which means that on every trading day, the optimization is run and weights of individual stock assets vary on a daily basis. In all subsequent analyses, configuration with $\lambda_T = 0.01$ is used.

Subsequently, it is necessary to define crucial evaluation metrics. Using these metrics, the comparison of portfolios with each other becomes possible and interpretable.

Suppose portfolio is applied on time interval $[1, N]$ and value of the portfolio is a time series $V = V_1, V_2, \ldots, V_N$

Cumulative annual growth rate, representing the average percentage return per year, is given as follows:

$$\mathrm{CAGR}(V) = \left(\frac{V_N}{V_1}\right)^{\frac{1}{r}} - 1, \tag{3}$$

where r denotes number of trading years between 1 and N timepoints.

Further, to quantify the risk and volatility of the portfolio **annualized volatility** (AV) can be adopted. Let $r_t = \frac{V_t}{V_{t-1}} - 1$ denote return on day t. Volatility across the given time interval is calculated as the sample standard deviation of $r_1, r_2, \ldots, r_N$ (denoted as σ). Subsequently, annualized volatility is given as follows:

$$\mathrm{AV}(V) = \sigma\sqrt{P}, \tag{4}$$

where P is the number of trading days per year.

Lastly, one compound metric, based on Equations (3) and (4), can be derived. **Sharpe ratio** represents the return a strategy earns per unit of risk and is given as:

$$\text{Sharpe ratio}(V) = \frac{CAGR(V)}{AV(V)} \tag{5}$$

5.2 Optimization on training-validation interval

Tuned models in Section 3 are used as a prediction engine for portfolio optimization. This means, that for particular trading day, models are employed to define the vector of expected returns $\boldsymbol{\mu}_t$. With obtained predictions, the only parameter of the optimization model to tune is λ_R, steering the risk-aversion.

In Figure 5, the dependence of CAGR on λ_R (in the Lasso General model) is depicted. It is possible to observe that larger λ_R (a more conservative portfolio) yields larger CAGR.

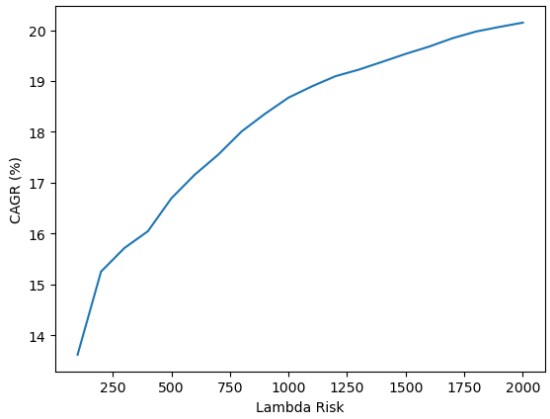

Figure 5: Dependence of portfolio CAGR on risk-aversion parameter (Lasso General model).

On top of that, in terms of CAGR dependence, a similar pattern was observed in all three analyzed model frameworks. Each portfolio, backed by predictions made by the particular model, exhibits growing returns with increasing values of λ_R.

Speaking of AV, monotonic course is observed as well. Figure 6 shows dependence of AV on the value of λ_R.

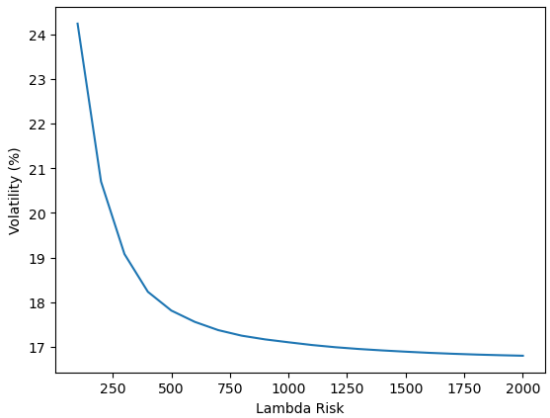

Figure 6: Dependence of portfolio AV on risk-aversion parameter (Lasso General model).

Even in this case, such declining course is observed in all three model frameworks.

Even though pattern is universal for all the frameworks, each implementation was configured differently, as the optimal value was analyzed independently.

In all the models, λ_R was chosen as the highest value beyond which improvement in both CAGR and AV becomes negligible. In cases of Lasso and Ridge, General model variant was evaluated as the differences between respective General and Stockwise portfolios were negligible. Overview of configurations is in Table 5.

Model	$\boldsymbol{\lambda_R}$
ARIMA	1000
Ridge	1200
Lasso	2000

Table 5: Portfolio model configuration overview.

The evaluation of results on the training-validation interval (as the simulation of investment) is presented in Table 6. The naïve strategy stands for portfolio optimization performed with naïve benchmark prediction model. As a benchmark, conventional stock indices NASDAQ and S&P 500 are considered as well.

Model	**CAGR (%)**	**AV (%)**	**Sharpe ratio**
Naïve	22.38	17.01	1.32
ARIMA	22.25	16.94	1.31
Ridge	21.67	16.96	1.25
Lasso	18.67	17.1	1.09
S&P 500	7.69	21.75	0.35
NASDAQ	9.19	25.52	0.36

Table 6: Evaluation of tuned models on training-validation interval.

Directly comparing results obtained from indices and models, portfolios backed by predictions performed marginally better in all reported metrics. On one hand, during the analyzed period, segment of the IT experienced vast rise, which is expected, but on the other hand, the NASDAQ index consists mainly of the technology and IT companies, so one could expect similar performance compared to the model-backed strategies [**?**].

This means, that model backed strategies are capable of being both less risky and better in terms of average annual return.

Overview of ARIMA backed strategy and investments in S&P 500 and NASDAQ are visualized in Figure 7.

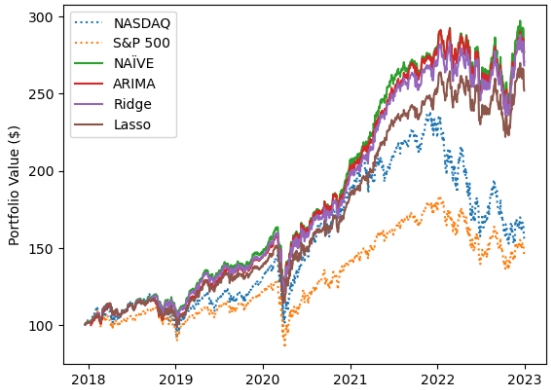

Figure 7: Overview of portfolio value across training-validation interval.

5.3 Test evaluation

Optimized portfolio strategies were applied to the test interval as well. The results are presented in Table 7, together with the performance of benchmark stock indices in the same time period.

Model	CAGR (%)	AV (%)	Sharpe ratio
Naïve	16.48	25.20	0.65
ARIMA	15.67	25.48	0.62
Ridge	15.29	25.24	0.6
Lasso	17.07	25.26	0.68
S&P 500	8.35	23.1	0.36
NASDAQ	18.51	25.15	0.74

Table 7: Evaluation of tuned models on test interval.

Diametrically different evaluations of the frameworks is observed. On the shorter test interval, benchmark indices outperformed all the model-backed strategies in all metrics. What is more, the models exhibited the opposite behavior on this interval. While on the training-validation interval, models with the highest dominance reported the best results, on the test interval quite the opposite was observed.

Paradoxically, the framework with the worst prediction results delivered the best-performing portfolio strategy. This is quite an interesting result in the context of the market regime in the time interval. During this period, the US market (particularly the IT and technology segment) experienced **bull market** – a phase in which the market in general experiences higher returns and profits [Graeber, 2023].

These results suggest that MVP-backed strategies performed poorly in a strong bull market phase. On the other hand, during the relatively mediocre phase between 2017 and 2022, model-backed strategies performed above expectations.

On top of that, these results suggest that employing the MVP with the proposed naïve prediction benchmark creates a resilient portfolio strategy also. Other model-backed strategies were unable to outperform the naïve portfolio strategy.

The final evaluation of all the employed models is in Figure 8.

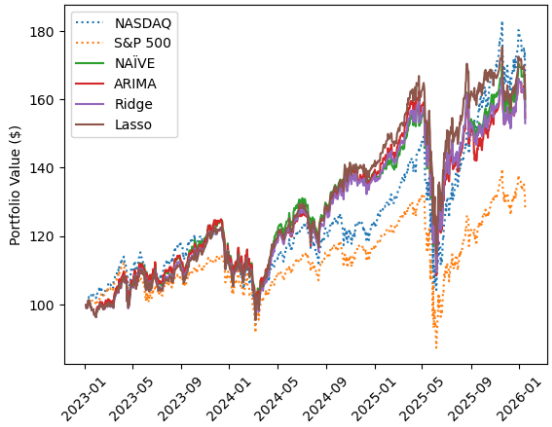

Figure 8: Overview of portfolio value across testing interval.

6 Conclusions

The results of both prediction modeling and portfolio optimization indicate that the predictability of the IT segment is relatively low. This finding aligns with Fama [Fama, 1970], who suggests that stock prices tend to follow a random-walk pattern.

Even with careful data preparation and feature engineering, conventional ML frameworks generally did not outperform the naïve benchmark. Many previous studies fail to compare predictions to a benchmark [Matsumoto et al., 2023, Yin et al., 2025], or compare models only among themselves [Krauss et al., 2017], which may overstate the statistical significance of their results.

Interestingly, the study demonstrates that building a robust and relatively profitable portfolio—compared to conventional indices—does not require precise and reliable market predictions. The employed MVP framework emphasizes the effect of the covariance-matrix quadratic form rather than the direct relationship between expected returns and portfolio weights [Palomar, 2025].

From a technical perspective, a key contribution of this work is the development of a modular data-pipeline engine. This pipeline allows the same methodology to be applied to any set of financial assets, facilitating studies of predictability and portfolio optimization across different market segments.

Finally, this framework may serve as a foundation for future research employing more sophisticated ML techniques, including advanced feature engineering, ensemble methods, deep learning architectures, or multi-horizon forecasting.

References

[Aroussi, 2025] Aroussi, R. (2025). yfinance: Yahoo! Finance market data downloader. Available at: `https://github.com/ranaroussi/yfinance`. Accessed: 2025-09-29.

[Bank for International Settlements, 2025] Bank for International Settlements (2025). OTC foreign exchange turnover in April 2025. Statistical release, Bank for International Settlements. Preliminary results; Final data released December 2025.

[Bergmann, 2025] Bergmann, D. (2025). What is machine learning? Available at: `https://www.ibm.com/think/topics/machine-learning`. Accessed: 2025-02-01.

[Diebold, 2006] Diebold, F. X. (2006). *Elements of Forecasting*. Thomson South-Western, 4th edition.

[et al., 2024] et al., N. C. (2024). 2024 Dividend Trends: A Global Overview. Visual Capitalist Elements. Accessed: 2025-12-04.

[Fama, 1970] Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. *Journal of Finance*, 25(2):383–417.

[Graeber, 2023] Graeber, D. J. (2023). S&P 500 in formal bull market territory, up 20% from October levels. United Press International (UPI). Accessed: 2026-02-27.

[Grudniewicz and Ślepaczuk, 2023] Grudniewicz, J. and Ślepaczuk, R. (2023). Application of machine learning in algorithmic investment strategies on global stock markets. *Research in International Business and Finance*, 66:102045.

[Huang, 2024] Huang, Y. (2024). Research on the Google Stock Price Prediction Based on SVR, Random Forest, and KNN Models. *Highlights in Business, Economics and Management*, 24:1054–1058.

[James et al., 2023] James, G., Witten, D., Hastie, T., Tibshirani, R., and Taylor, J. (2023). *An Introduction to Statistical Learning*. Springer Texts in Statistics. Springer Nature.

[Kenton, 2025] Kenton, W. (2025). S&P 500 Index: What It's for and Why It's Important in Investing. Accessed: 2025-12-03.

[Khaidem et al., 2017] Khaidem, L., Saha, S., and Dey, S. R. (2017). Predicting the direction of stock market prices using random forest. *Expert Systems with Applications*, 83:86–99.

[Krauss et al., 2017] Krauss, C., Do, X., and Huck, N. (2017). Deep neural networks, gradient-boosted trees, random forests: Statistical arbitrage on the S&P 500. *European Journal of Operational Research*, 259(2):496–502.

[Kumar et al., 2021] Kumar, R. R., Stauvermann, P. J., and Vu, H. T. T. (2021). The Relationship between

Yield Curve and Economic Activity: An Analysis of G7 Countries. *Journal of Risk and Financial Management*, 14(12):574.

[Mantilla and Dormido-Canto, 2023] Mantilla, P. and Dormido-Canto, S. (2023). A novel feature engineering approach for high-frequency financial data. *Engineering Applications of Artificial Intelligence*, 125:106394.

[Markets, 2025] Markets, C. (2025). What is currency trading? CMC Markets. Accessed: 2025-11-29.

[Matsumoto et al., 2023] Matsumoto, A., Pescatori, A., and Wang, X. (2023). Commodity prices and global economic activity. *Japan and the World Economy*, 66:101117.

[Morabito, 2025] Morabito, C. (2025). The S&P 500 is more concentrated with AI than ever. Here's how to manage your risk. Accessed: 2025-12-03.

[Osita, 2025] Osita, J. (2025). Why Gold Remains the Ultimate Security in a Shifting World. ACY Securities. Accessed: 2025-11-27.

[Palomar, 2025] Palomar, D. P. (2025). *Portfolio Optimization: Theory and Application*. Cambridge University Press.

[Sakina and Khan, 2024] Sakina and Khan, M. A. (2024). Stock Price Prediction using Neural Networks. *Pakistan Journal of Law, Analysis and Wisdom*, 3:123–135.

[Sezer et al., 2020] Sezer, O. B., Gudelek, U. M., and Ozbayoglu, A. M. (2020). Financial Time Series Forecasting with Deep Learning : A Systematic Literature Review: 2005-2019. *Applied Soft Computing*, 90:106181.

[Skiena, 2017] Skiena, S. S. (2017). *The Data Science Design Manual*. Texts in Computer Science. Springer Nature.

[Smith, 2026] Smith, T. G. (2026). pmdarima. Available at: `https://github.com/alkaline-ml/pmdarima`. Accessed: 2026-02-23.

[Wang et al., 2023] Wang, Q., Xu, C., and Zhou, T. (2023). Stock price prediction based on multiple linear regression. *BCP Business & Management*, 36:48–54.

[Yahoo Finance, 2025] Yahoo Finance (2025). Yahoo Finance. Available at: `https://finance.yahoo.com/`. Accessed: 2025-09-29.

[Yin et al., 2025] Yin, L., Zhang, X., and Xiong, X. (2025). Application and Effectiveness of KNN Model in Stock Market Prediction. In *Proceedings of the 3rd International Conference on Financial Technology and Business Analysis*, pages 129–133.

[Zhao, 2025] Zhao, Y. (2025). Stock Price Prediction Based on Linear Regression. In *Proceedings of the 4th International Conference on Computing Innovation and Applied Physics*, EAI/Springer Innovations in Communication and Computing, pages 85–90. EAI.